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$$f(x) = (\ln x)^x$$

$$f(x) = e^{x \ln(\ln x)}$$

$$f'(x) = e^{x \ln(\ln x)} (x \ln(\ln x))'$$

$$= \ln(\ln x) + 1/(\ln x)$$

$$f'(x) = (\ln x)^x (\ln(\ln x) + 1/(\ln x))$$

$$= (\ln x)^x \ln(\ln x) + (\ln x)^x (1/(\ln x))$$

$$= (\ln x)^x \ln(\ln x) + (\ln x)^{x-1}$$

$$\boxed{f'(x) = (\ln x)^{x-1} + (\ln x)^x \ln(\ln x)}$$

References